

nomadAI

AI-driven dynamic index reweighting for asset managers and ETF providers.

nomad team

trading & investing

Centennial Campus, HKU • 4 October 2026

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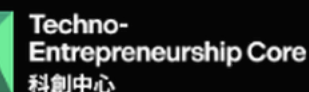
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Partners



Supporting organisation



▪ [Team pitch // Live demo]

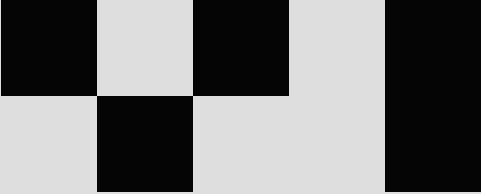
▪ iFX Hack Hong Kong

▪ Theme: Build the Future of Finance with AI

Team Nomad

Members Yutaka, Nurtilek, Temirlan, Sanzhar

Built with Amazon Bedrock, Python, scikit-learn

- 
- The Problem]

The problem

Market stress can increase portfolio risk as asset prices begin moving together.

Today they [current workaround], which costs them [time / money / risk].

- // EVIDENCE

[ADD SOURCE] Trillions of dollars are trapped in static indices that act too slowly to emerging contagion risk, exposing retail and institutional investors to avoidable drawdowns.

▪ The Challenge]

Where we play

Quantitative portfolio risk
management

HONG KONG

Tencent Holdings, Alibaba Group, Meituan, Xiaomi Corporation, JD.com, Cash

We also tested model in markets:

- SINGAPORE
- SHANGHAI

6% alpha generated (outperformed) vs.
HSI baseline during peak stress windows

[6 %]

[COST OF THE PROBLEM]

Our solution

nomadAI AI helps ETF providers **minimize drawdowns** by continuously recalculating risk using graph theory and live market sentiment.

01

Feature 1

A Random Forest model that analyzes volatility and momentum to predict market turbulence.

02

Feature 2

Mathematical graph that tracks correlations between assets, identifying which stocks that can crash multiple stocks along with them.

03

Feature 3

Automated risk manager that scales down asset allocation and redirects capital to cash when extreme scenario is detected.

How the AI works

STEP 1

▪ DATA IN

Live market headlines
and historical ticker
data.

STEP 2

▪ AI LAYER

Amazon Bedrock evaluates
headline risk sentiment;
a Random Forest
regressor computes
dynamic feature

STEP 3

▪ OUTPUT

Constrained linear
algebra algorithm
generates the final
optimized index weights.

STEP 4

▪ VALUE

Reduced portfolio
volatility and active
risk mitigation.

// BUILT WITH:

[AWS service, e.g. Amazon Bedrock]

[AWS service]

[Model / framework]

[Data / API]

Business model

WHO PAYS	WHAT THEY GET	HOW WE CHARGE
Index providers & ETF issuers	Dynamic risk-adjusted index methodology	Basis point (bps) AUM licensing fee
Pension funds	Downside protection and asset rebalancing	SaaS fee or AUM basis points
Investment banks	Quantitative risk and optimization API	API usage fee or enterprise license

What's next

Data Collection

STEP 1

To get more possibilities for other markets, we need more data and tokens to upload it in real life.

▪ BEYOND THE HACK

NEXT

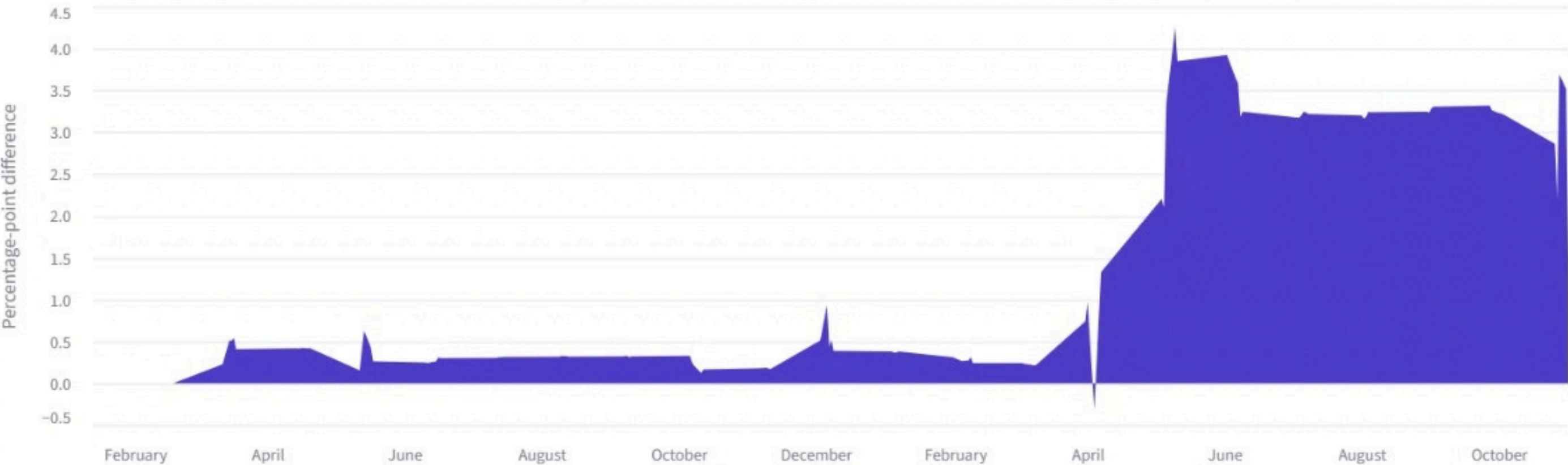
6 MONTHS

Launch a paper-traded pilot index for institutional peer review

Backtest date range
2020-06-19 2022-11-19

Available history: 2020-06-19 to 2026-09-30

Basket companies: Tencent Holdings, Alibaba Group, Meituan, Xiaomi Corporation, JD.com, and Cash



Proof it works

[3]

Biggest financial
cities in Asia

[4%]

Difference profited from equal-weight
distribution

[7]

research papers analyzed,
including:
Bae, Kim & Mulvey (2014),
"Dynamic asset allocation for
varied financial markets under
regime switching"

Measured on [date] using [test data / real users].

▪ Our Ask]

BIGGER COMPUTER POWER TO TRAIN BETTER MODELS

which will prove significant results

// CONTACT

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// DEMO / CODE

github.com/yutakatarokusu
mah/ifx-hack

// TEAM

nomad.

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Partners



Supporting organisation



▪ nomadAI

QA & the team



Tim

fin analyze

@ MIT media lab
@ HKUST x CAEA



yutaka

ai

@ INDONESIA Math
team



sanzhar

math, cs

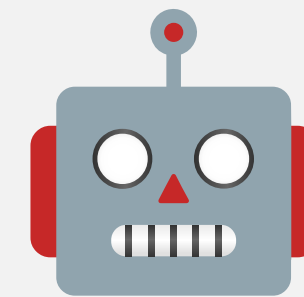
@ SDU Econ
research



nurtilek

ai

@ Columbia
research



our model - nomad

6% outperformed
HKEX market

Our mix: [technical] · [ai] · [product] · [finance]